



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma Yellow Belt (CLSSYB) Training

TGS Ref No: TGS-2025053922

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 8

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release — CLSSYB 2-day lesson plan aligned to the 10 hands-on labs.	Dr. Alfred Ang
2	15 June 2026	Refreshed labs against the CSSC Yellow Belt reference.	Dr. Alfred Ang
3	1 July 2026	Aligned the schedule with the Contoso Service Desk case-study assessment.	Dr. Alfred Ang
4	23 September 2026	Major content expansion: course restructured to follow the DMAIC roadmap end to end; Lean Six Sigma concepts imported from the reference trainer deck and the CSSC body of knowledge and explained visually; 4 elective labs added (Affinity/Kano, VSM/Takt, Solution Matrix/Benchmarking/FMEA, Descriptive Statistics/Implementation Plan) and Lab 4 (PDCA charter) reclassified as elective, giving 9 core and 5 elective labs; interactive problem-solving tools (5 Whys, Fishbone, collaborative Pareto, NovaSPC) integrated into the Analyze labs; slide references added to the schedule.	Dr. Alfred Ang
5	23 September 2026	Integrated the SIPOC & Process Map Builder (alfredang.github.io/sipoc/) into Lab 3 and elective Lab 12, with a tool walkthrough added to the Define phase and the online toolkit slide updated to list all five tools.	Dr. Alfred Ang
6	23 September 2026	Fixed clipped headings: slide titles now auto-fit to a single line so long lab titles	Dr. Alfred Ang

		can no longer overprint the LAB/ELECTIVE chips beneath them.	
7	20 July 2026	Table of contents restored to a live, updatable Word field (previously flattened to static text), so it refreshes on open or F9 while still rendering correctly in the distributed PDF.	Dr. Alfred Ang
8	23 September 2026	Labs restructured into one self-contained folder per lab, each with its own mock data (data/*.csv) and blank worksheets (templates/*.csv). The Contoso Service Desk data set is now generated from a single verified source and reconciles exactly with the Case Study assessment figures (400 tickets, 96 defective, 120 defects, yield 76%, DPU 0.30, DPMO 50,000). Slides, Lesson Plan and Learner Guide updated to reference the lab data.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information.....	5
Learning Outcomes.....	5
Training Resources and Lab Data.....	5
Assessment.....	6
Course Schedule.....	6
Day 1 — Six Sigma foundations, Define phase and process mapping.....	6
Day 2 — Measure, Analyse, Improve, Control and assessment.....	7
Lab Reference (aligned to the DMAIC phases).....	8

Course Information

Course Title	Certified Lean Six Sigma Yellow Belt (CLSSYB) Training
WSQ Course Reference	TGS-2025053922
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the Contoso Service Desk improvement scenario
TSC Alignment	Quality Process Control (ELE-QUA-5006-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Define a project and establish the scope of work using Six Sigma's Define, Measure and Analyse phases.
- LO2: Apply Lean and Six Sigma concepts — value, waste, defects and variation — to a work process.
- LO3: Map a process using SIPOC, process maps and value stream maps to expose handoffs and waste.
- LO4: Collect and analyse process data using check sheets, Pareto charts, run charts and basic metrics.
- LO5: Identify root causes using 5 Whys, Fishbone analysis and evidence-based prioritisation.
- LO6: Recommend improvement and control actions to sustain gains, and prepare for certification.

Training Resources and Lab Data

Each lab is a self-contained folder under labs/ — labs/lab-NN-<name>/ — holding the worksheet (README.md), the mock data the learners analyse (data/*.csv) and the blank worksheets they complete (templates/*.csv).

- Shared data set: two weeks of Contoso Service Desk activity — 400 tickets, 96 defective, 120 defects, 6 opportunities per ticket.
- Baseline performance: yield 76%, DPU 0.30, DPMO 50,000, sigma level ~3.1; mean assignment time 55 min (median 46) against a 30-minute goal.

- These are the same figures as the Case Study assessment, so lab work is direct revision.
- Trainer note: the Lab 7 worked answers are in labs/lab-07-*/solution/ — do not release before learners attempt it.
- Learners need a spreadsheet application (Excel, Google Sheets or LibreOffice Calc) and a browser.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 60 minutes, open book.
- Case Study (CS) — applied Lean Six Sigma tasks, 90 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Six Sigma foundations, Define phase and process mapping

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM) [slides 2–15]
10:00–11:15	75 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean vs Six Sigma vs Lean Six Sigma; sigma levels and DPMO; Y = f(X); belt roles; the DMAIC roadmap; DMAIC vs PDCA [slides 16–39]
11:15–11:30	15 min	Tea break (within training time)
11:30–12:30	60 min	Hands-on: Lab 1: Yellow Belt Role, Certification Paths, and Improvement Scenario [slides 40–47]
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; the project charter; problem and goal statements; scope; SIPOC; process mapping levels and symbols [slides 48–67]
15:00–15:15	15 min	Tea break (within training time)
15:15–18:00	165 min	Hands-on (core): Lab 2: Lean, Six Sigma, Waste, Voice of Customer, and Value; Lab 3: SIPOC, Process Mapping,

		Handoffs, and SME Support; Lab 5: DMAIC Overview, Problem Statement, Scope, and Stakeholders. Elective (time permitting): Lab 4: Elective — PDCA Small Improvement Project Charter; Lab 11: Elective — Affinity Diagram and Kano Analysis [slides 68–111]
18:00–18:30	30 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours) — includes tea breaks, excludes the 1-hour lunch.

Day 2 — Measure, Analyse, Improve, Control and assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	DMAIC · MEASURE — the eight wastes (DOWNTIME); value-added analysis; types of data; data collection plans and operational definitions; check sheets; sampling; yield, DPU, DPO, DPMO; sigma level; COPQ [slides 112–132]
11:00–11:15	15 min	Tea break (within training time)
11:15–12:30	75 min	Hands-on (core): Lab 6: Data Collection, KPIs, Check Sheets, and Basic Metrics. Elective (time permitting): Lab 12: Elective — Value Stream Map and Takt Time [slides 133–149]
12:30–13:30	60 min	Lunch break
13:30–14:15	45 min	DMAIC · ANALYZE — common vs special cause variation; Pareto principle and charts; run charts and pattern rules; 5 Whys; Fishbone (5M); brainstorming and multi-voting; the interactive problem-solving toolkit [slides 150–171]
14:15–15:00	45 min	Hands-on (core): Lab 7: Pareto, Run Charts, Variation, Yield, DPU, and DPMO; Lab 8: Root Cause Analysis with 5 Whys, Fishbone, and Evidence using the collaborative Pareto, 5 Whys, Fishbone and NovaSPC tools [slides 172–193]
15:00–15:15	15 min	Tea break (within training time)
15:15–15:30	15 min	DMAIC · IMPROVE — solution generation and

		selection; 5S; poka-yoke; standard work; Kaizen; piloting [slides 194–208]. Hands-on (core): Lab 9: Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen. Elective: Lab 13: Elective — Solution Selection Matrix, Benchmarking, and FMEA [slides 209–227]
15:30–15:45	15 min	DMAIC · CONTROL — control plans; process control; visual management; SOPs; team huddles; Gemba; A3 and handover [slides 228–242]. Hands-on (core): Lab 10: Control Plan, A3 Summary, Handover, and Certification Readiness. Elective: Lab 14: Elective — Descriptive Statistics and Implementation Planning [slides 243–259]
15:45–16:00	15 min	Course recap and Briefing for Assessment [slides 260–270]
16:00–17:00	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 60 minutes, open book
17:00–18:30	90 min	Case Study (CS) — applied Lean Six Sigma tasks, 90 minutes, open book. PM digital attendance

Total training time: 480 minutes (8 hours) — includes tea breaks, excludes the 1-hour lunch.

Lab Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Labs
FOUNDATIONS: Six Sigma Foundations	15%	Lab 1
DEFINE: Define — Scope the Problem	25%	Lab 2, Lab 3, Lab 4 (elective), Lab 5, Lab 11 (elective)
MEASURE: Measure — Quantify Performance	25%	Lab 6, Lab 12 (elective)
ANALYZE: Analyze — Find the Root Cause	20%	Lab 7, Lab 8
IMPROVE: Improve — Fix the Cause	10%	Lab 9, Lab 13 (elective)
CONTROL: Control — Hold the Gain	5%	Lab 10, Lab 14 (elective)